

SIMATS ENGINEERING
ASSESSMENT TOOL 3: Short Answer Test

COURSE NAME: Cloud Computing and
Big Data Analytics

COURSE CODE: CSA1522

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COURSE OUTCOME COVERED

CO1: Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions. (BL2)

Assessment Tool Weightage: 40%

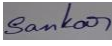
Dave's Taxonomy: Imitation (Level 1)
Question Count: 5

Total Marks: 50

Code of Conduct

I Kadiri Sankar Narayana (192325085 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: 

Assessment Tasks

Short Answer Test

1. Define cloud computing and explain any four characteristics that make it different from traditional on-premise computing.
2. Trace the evolution of cloud computing from hardware evolution to internet software evolution, and show how these changes enabled modern cloud platforms.
3. Explain the role of server virtualization in cloud computing. Differentiate between virtualization as an enabling technology and cloud computing as a service model.
4. Compare SOAP-based and REST-based web services in terms of communication style, data exchange, and suitability for cloud applications.
5. Differentiate IaaS, PaaS, SaaS, and XaaS with one suitable real-world example for each model.

Short Answer Test Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Conceptual Accuracy	30%	Accurate definitions and precise explanation of all key terms	Mostly accurate with minor gaps	Partial understanding with noticeable errors	Incorrect or irrelevant explanation
Coverage of Concepts	20%	Covers all expected sub-points completely	Covers most expected points	Covers only basic points	Very limited coverage

Use of Examples	15%	Uses highly relevant cloud examples to strengthen the answer	Uses relevant examples with minor mismatch	Uses vague or partially relevant examples	No useful examples
Comparison / Differentiation	20%	Clearly distinguishes concepts with strong logic	Reasonable differentiation with minor overlap	Comparison is weak or incomplete	Fails to differentiate concepts
Clarity of Expression	15%	Clear, organized, and concise answer writing	Generally clear with few issues	Understandable but poorly organized	Difficult to follow

1. Define Cloud Computing and Explain Any Four Characteristics

Cloud Computing:

Cloud computing means using computers, storage, software, and other services through the internet instead of using your own computer or server.

Characteristics:

On-demand service – Users can use cloud services whenever they need.

Access from anywhere – Cloud services can be used from any place with an internet connection.

Scalability – Resources can be increased or decreased based on need.

Pay as you use – Users pay only for the services they use.

2. Evolution of Cloud Computing

Hardware Evolution

Early computers were large mainframes shared by many users.

Personal computers later enabled individual computing.

Powerful servers and virtualization improved hardware utilization.

Internet Evolution

High-speed internet enabled remote access to computing resources.

Distributed computing allowed systems to communicate over networks.

Software Evolution

Applications evolved from desktop software to web-based applications.

Service-Oriented Architecture (SOA), Web Services, and APIs enabled software integration.

Cloud Computing Emergence

Virtualization + High-speed Internet + Web Services + Data Centers resulted in cloud computing.

Providers like AWS, Microsoft Azure, and Google Cloud deliver scalable computing resources on demand.

Evolution Flow:



3. Role of Server Virtualization in Cloud Computing

Server Virtualization

Server virtualization divides one physical server into multiple virtual machines (VMs), each running its own operating system independently.

Role in Cloud Computing

Improves hardware utilization.

Reduces infrastructure cost.

Enables rapid deployment of servers.

Supports scalability and resource isolation.

Simplifies backup and disaster recovery.

Virtualization	Cloud Computing
Technology that creates virtual machines	Service model delivering IT resources over the internet
Focuses on efficient hardware utilization	Focuses on providing services to users
Runs on a single physical server	Uses virtualization plus networking, automation, and management
Enables cloud infrastructure	Uses virtualization as one of its core technologies

4. Difference Between SOAP and REST Web Services

Feature	SOAP	REST
Full Form	Simple Object Access Protocol	Representational State Transfer
Communication Style	Protocol-based	Architectural style
Data Format	XML only	JSON, XML, HTML, Text
Performance	Slower	Faster
Complexity	More complex	Simpler
Security	Built-in WS-Security	Uses HTTPS and OAuth
Best Suitable For	Banking, Enterprise applications	Cloud, Mobile, Web APIs

. 5. Differentiate IaaS, PaaS, SaaS, and XaaS

Service Model	Description	User Manages	Example
IaaS (Infrastructure as a Service)	Provides virtual servers, storage, and networking	OS, applications, data	Amazon EC2
PaaS (Platform as a Service)	Provides platform for application development	Applications and data	Google App Engine
SaaS (Software as a Service)	Provides ready-to-use software over the internet	Only user data	Gmail or Microsoft 365

Service Model	Description	User Manages	Example
XaaS (Anything as a Service)	Any IT service delivered over the internet	Depends on service	AWS (multiple cloud services), Dropbox Storage-as-a-Service

Summary

- **IaaS:** Rent infrastructure.
- **PaaS:** Rent a development platform.
- **SaaS:** Use software directly.
- **XaaS:** Any service delivered through the cloud.